

Atmospheric Scattering

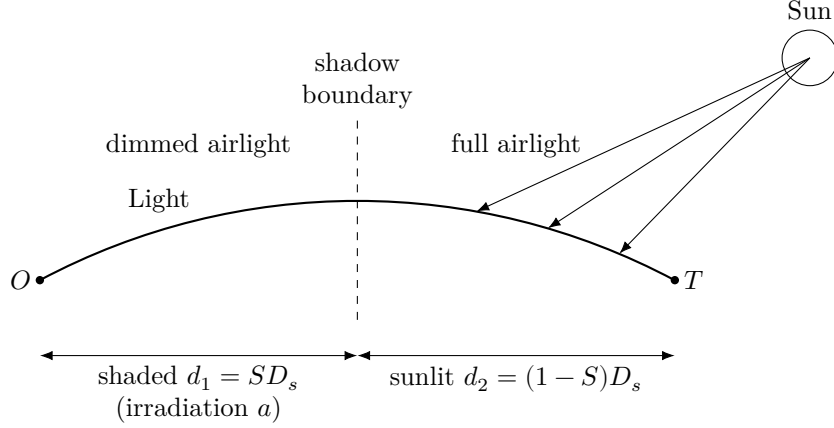
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A geometrically unobstructed line of sight is not enough for the target to be visible: air scatters light, so a distant target fades into the background sky. This document assembles the **contrast** of the target against the sky, following Michael Vollmer, *Below the horizon — the physics of extreme visual ranges* (2020). The target is considered visible when its contrast exceeds 0.02, the threshold of human vision. This is computed in `LineOfSight.get_contrast` and `LineOfSight.has_contrast` in `scripts/commons.py`.

In polar coordinates the observer is at $P_1 = (R_\oplus + h_1, 0)$ and the target at $P_2 = (R_\oplus + h_2, D_s/R_\oplus)$. Light follows the arc of a circle of radius $R_L = kR_\oplus$ (see `atmospheric_refraction.md`).

Notation

- C — contrast of the target against the sky.
 - h_1, h_2 — elevation of the observer and target.
 - D_s — surface (sea-level arc) distance between observer and target.
 - D_d — direct (straight-line) distance between observer and target.
 - H — scale height of the atmosphere.
 - β_0 — scattering coefficient at sea level.
 - $\beta(h) = \beta_0 e^{-h/H}$ — scattering coefficient at elevation h .
 - β_1, β_2 — average scattering coefficient over the shaded and sunlit segments.
 - S — shaded ratio: fraction of the path in shadow, starting at the observer.
 - a — shade irradiation ratio: brightness of the shaded segment relative to the sunlit segment.
 - x — surface distance from the observer; $\theta = x/R_\oplus$, $\phi = D_s/R_\oplus$.
 - $L(x)$ — distance from the centre of the Earth to the light arc at x .
 - $h(x) = L(x) - R_\oplus$ — height of the light arc above sea level at x .
 - (M_x, M_y) — centre of the light-arc circle in Cartesian coordinates.
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Partial irradiation: near sunrise/sunset the observer-side half of the path lies in shadow (dimmed airlight, irradiation ratio a), while the target-side half is still sunlit. Shadowing the near air, where airlight is strongest, is what lets extreme sightlines keep contrast.

Contrast (Vollmer eq. 9). The path is split into a shaded segment of length $d_1 = SD_s$ (starting at the observer) and a sunlit segment of length $d_2 = (1 - S)D_s$:

$$C = \frac{e^{-\beta_2 d_2}}{1 - a + \left(\frac{a}{e^{-\beta_1 d_1}} \right)}$$

Average scattering coefficients (Vollmer eq. 7). Scattering thins with altitude, so the sea-level coefficient is weighted by the height of the light path and integrated along it. The integrals run over the light path, with arc-length element ds :

$$\beta_1 = \frac{\beta_0}{d_1} \int_0^{d_1} e^{-h(x)/H} ds, \quad \beta_2 = \frac{\beta_0}{d_2} \int_{d_1}^{D_s} e^{-h(x)/H} ds$$

$$d_1 = SD_s, \quad d_2 = (1 - S)D_s, \quad h(x) = L(x) - R_\oplus$$

Here ds is the arc-length element of the light path. Parameterising by surface distance x (so $\theta = x/R_\oplus$), $ds = \sqrt{(L/R_\oplus)^2 + (dL/dx)^2} dx \approx (L/R_\oplus) dx$, so in code the surface-distance integral is weighted by $L/R_\oplus$ (see `LineOfSight.get_contrast`).

Height of the light arc. With $\theta = x/R_\oplus$, the light circle in polar form is

$$L(x) = M_x \cos \theta + M_y \sin \theta + \sqrt{(M_x \cos \theta + M_y \sin \theta)^2 + R_L^2 - M_x^2 - M_y^2}$$

where the centre of the light circle and the endpoints are

$$M_x = \frac{x_1 + x_2}{2} - Z \frac{y_2 - y_1}{D_d}, \quad M_y = \frac{y_1 + y_2}{2} + Z \frac{x_2 - x_1}{D_d}, \quad Z = \sqrt{R_L^2 - \left(\frac{D_d}{2}\right)^2}$$

$$P_1 = (x_1, y_1) = (R_\oplus + h_1, 0), \quad P_2 = (x_2, y_2) = ((R_\oplus + h_2) \cos \phi, (R_\oplus + h_2) \sin \phi)$$

$$D_d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, \quad \phi = \frac{D_s}{R_\oplus}$$

Choosing S and a . The shaded segment is the near, observer-side half of the path ($S = 0.5$), because airlight is dominated by the air closest to the observer, so shadowing it helps contrast most. The shade irradiation ratio a is set per line of sight from its bearing, to approximate sunrise/sunset. The sun can only rise or set along the east-west horizon, so only an east-west sightline can place the observer's foreground in shadow with the target lit beyond; a north-south sightline has the sun broadside and gets no benefit. Hence $a = 1 - (1 - a_{\min}) |\sin(\text{bearing})|$ with $a_{\min} = 0.1$: east-west gives $a = 0.1$ (deepest shadow, highest contrast), north-south gives $a = 1$ (plain extinction), and bearings between interpolate. $a_{\min}$ stays above 0 because fully shadowed air is still lit by diffuse skylight.

Irradiation by bearing. The table below evaluates $a = 1 - (1 - a_{\min}) |\sin(\text{bearing})|$ at 15° steps, alongside the resulting reduction in shaded-segment irradiation, $1 - a = (1 - a_{\min}) |\sin(\text{bearing})|$. Because the relation depends on $|\sin(\text{bearing})|$, it is symmetric every 180° : the reduction is largest ($1 - a = 0.9$) for east-west sightlines and vanishes for north-south ones.

Bearing	Reduction ($1 - a$)	Irradiation (a)
0° (N)	0.000	1.000
15°	0.233	0.767
30°	0.450	0.550
45°	0.636	0.364
60°	0.779	0.221
75°	0.869	0.131
90° (E)	0.900	0.100
105°	0.869	0.131
120°	0.779	0.221
135°	0.636	0.364
150°	0.450	0.550
165°	0.233	0.767
180° (S)	0.000	1.000
195°	0.233	0.767
210°	0.450	0.550
225°	0.636	0.364

Bearing	Reduction ($1 - a$)	Irradiation (a)
240°	0.779	0.221
255°	0.869	0.131
270° (W)	0.900	0.100
285°	0.869	0.131
300°	0.779	0.221
315°	0.636	0.364
330°	0.450	0.550
345°	0.233	0.767
360° (N)	0.000	1.000